A kinetic-impact deflection model

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to explain your model or result. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

impulse: net force acting over time; change in momentum

01 Bring back an earlier idea

Two equal 3 kg carts approach at 5 m/s in opposite directions and stick.
Find their common final velocity and explain.

Word help: final = at the end of the interval; subscript f

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × time.
Toy model: much heavier target; straight-line shift.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

A tiny sideways velocity change acts for a short time or a long time. Which produces more sideways shift in the straight-line model?
Explain.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

04 Build the solution

A supplied target velocity change is .01 m/s sideways.
Calculate shift after 100 s and 1000 s.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

An asteroid’s period is 6 years.
Explain why that number alone cannot establish an Earth encounter.
Name two additional observations needed.

Word help: encounter = objects reaching nearby positions at the same time

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

At the same smallΔv, compare shift with twice as much lead time.
State one model limit.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Keep projectile momentum 60 kg·m/s but double the stationary target mass from 10 to 20 kg.
Calculate the new final speed and explain the change.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3